

Solid Mechanics

FEA MODELLING PROJECT



Submitted By:

JOYAL JOJO STUDENT ID: 1536734	RYAN HEWITT STUDENT ID: 1172261
-----------------------------------	------------------------------------

Executive Summary	2
Theoretical Analysis Part 1.....	3
a. <i>Determination of stresses and free body diagram</i>	3
b. Calculation of Stresses.....	4
c. Comparison between the hoop and longitudinal stress.....	9
d. Empty and hydrostatic condition analysis Empty vessel	10
e. Minimum Acceptable Thickness of skirt.....	13
f. Minimum acceptable thickness of the shell, top and bottom heads analysis.	14
g. Vibration Analysis.....	15
h. Role of Stress-Relieving	16
i. Thermal Stress	17
j. Initial thickness variation analysis	18
Modelling Analysis Part 1	18
a. Meshing and Element Type Selection	18
b. Wind/Seismic Load Analysis.....	19
c. Safety, Lifetime, and Recommendations	21
Practical Recommendations	22
Part 2: Analytical Study	23
a. Yield Pressure for Thin-Walled Vessel	23
b. Yield Pressure for Non-Thin-Walled-Vessel.....	23
c. Pressure to Yield Vessel to 5mm.....	24
d. Residual Stress Determination	25
e. Hoop Stress Calculation on Venus	26
Part 2: Numerical Study	27
a. Meshing and Stress Analysis with Tetrahedral Elements.....	27
b. Analysis using Shell Elements	27
c. Elasto-Plastic Material Analysis	28
d. Inclusion of Gravitational Effects.....	29
e. Evaluation of Support Effects and Design.....	30

Executive Summary

This report evaluates the integrity and safety of the pressure vessel V2-1979. Based on the analysis, the vessel is considered safe for continued operation under current conditions, though with a reduced safety margin. The expected remaining lifetime of the vessel is projected to be more than 15 years, assuming a uniform corrosion rate of 0.1 mm per year.

The key findings from the analysis include.

- **Minimum Thickness Requirement:** The minimum acceptable thickness of the skirt is determined to be within 3.7 mm to 4.1mm.
- **Stress Analysis:** Calculations of hoop and longitudinal stresses at various vessel sections demonstrate that the stresses are within acceptable limits.
- **Safety Factor:** The calculated safety factor for critical components of the vessel, such as the skirt and shell, is close to 1.6. Corrosion treatment and other strengthening operations in week spot can further increase the factor of safety to 2.
- **Thermal and Wind Effects:** The vessel is vulnerable to thermal expansion and wind loads, which have been accounted for in the stress analysis, showing that the vessel can handle expected variations without critical failure.
- **Vibration and Buckling:** The vessel is operating under critical Vibration and Buckling load.

Recommendations:

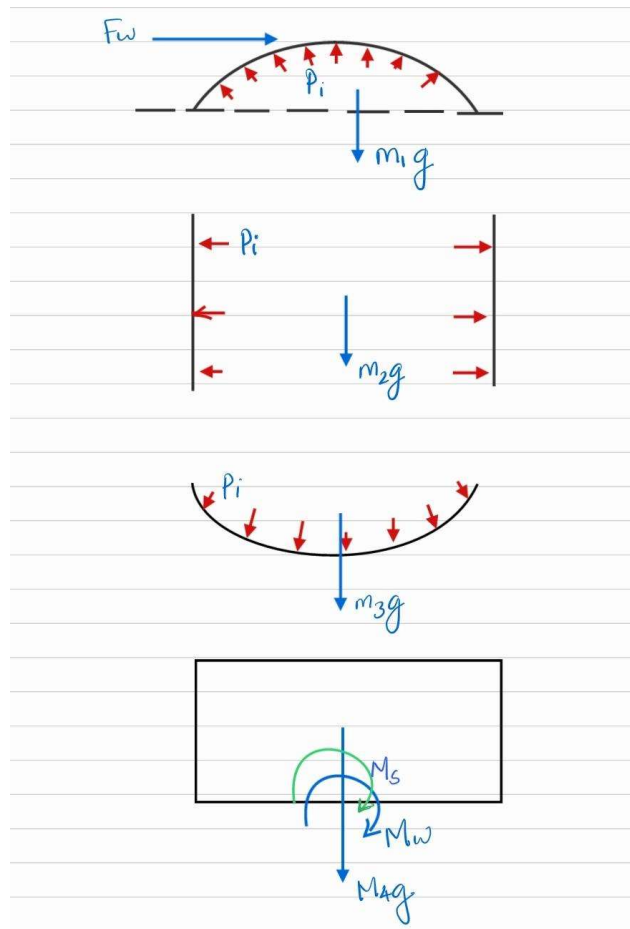
1. **Increased Inspection Frequency:** Given the criticality of the vessel, frequent inspections are recommended, particularly focusing on corrosion and stress concentration points such as the skirt and manhole areas.
2. **Corrosion Prevention:** Implementing corrosion-resistant coatings or regular maintenance to mitigate thickness loss from corrosion.
3. **Fatigue Analysis:** Conduct fatigue life estimation due to cyclic loading and thermal fluctuations to ensure long-term integrity.
4. **Retrofitting:** Consider reinforcing the skirt where thickness is between 3.5 to 9 mm to increase safety margins and extend the vessel's operational life.

Sources of Error:

- **Measurement Errors:** Assumed thickness of 13mm average skirt thickness could be in higher side if the laser scan data collected is wrong.
- **Residual Stresses:** The absence of stress-relieving during manufacturing introduces residual stresses that could not be fully quantified without non-destructive testing.

Theoretical Analysis Part 1

a. Determination of stresses and free body diagram



F_w	Wind Force (N)
M_1	Mass of Fluid + Ellipsoid head (Kg)
M_2	Mass of Fluid + Shell (Kg)
M_3	Mass of Fluid + Ellipsoid bottom (Kg)
M_4	Mass of Skirt + Mass of vessel (Kg)
M_s	Moment due to seismic activity (N-m)
M_w	Moment due to wind load (N-m)
P_i	Internal Pressure (Pa)

b. Calculation of Stresses

Assumptions: Thin wall Theory $\left(\frac{Radius}{Thickness}\right) \leq 10$

Thickness of the skirt is assumed to be 13mm. Even though there are places with less thickness than 13mm, those areas can be treated to increase thickness and selecting the minimum thickness 3.46mm doesn't mimic the actual stress distribution in the vessel.

Working formulae without Rotation:

$$\sigma_l = \frac{PR}{2t \cos \theta_l} + \frac{W}{2\pi R t \cos \theta_l} \quad (1)$$

$$\frac{d(\sigma_l R)}{dS_l} = \sigma_h \sin \theta_l \quad (2)$$

$$\frac{\sigma_l}{r_l} + \frac{\sigma_h}{r_h} = \frac{P}{t} \quad (3)$$

Wind Load and Moment

- Working life: 25 years. — AS 1170
- Importance level (IL): 4 — Page 381 (AS 1170)
- Region A5 (Geelong) — AS-NZS 1170
- (Why A5? Because collapsing a pressure vessel could cause the post operation of the plant to stop)
- So, for IL of 4 → Wind Annual probability of the design wind: 1/2000
- $P = (\text{Probability} \times 50) / N \rightarrow P = 1/2000 \times 50/25 = 1/1000$
- Annual probability of occurrence $P = 1/1000$
- Site Wind Speed:
- $V_{1000} = 46 \text{ m/s}$
- $V_R = 67 - 41R^{(-0.1)} = 46 \text{ m/s}$
- $P_w = 0.00256 \times V^2 \text{ psf} = 0.613 \times V^2 \text{ Pa} = 0.613 \times 46^2 = 1297 \text{ Pa} \rightarrow 27.1 \text{ psf}$
(US Civil Engineer Code)

Projected Area of Tower:

- $A_A: 30 \times 4.27 \text{ m}^2 = 128.1 \text{ m}^2$ (shell)
- $A_B: 4.2 \times (1.9 \times 2) \text{ m}^2 = 19.96 \text{ m}^2$ (skirt)
- $A_C: (\pi \times 1.1 \times 2.135) / 2 = 3.69 \text{ m}^2$ (head)

Force (F_w):

$$F_w = P_w \times G \times C_f \times A_f = 1297 \text{ Pa} \times 4.13 \times 0.8 \times (\text{cylinder area}) \Rightarrow \mathbf{F_w = 199.3 \text{ kN}}$$

Moment at the base of the tower = $199300 \times 35.3 = 7035.290 \text{ kN-m}$

AS 1270 Wind speed calculation

$$V_{des} = 0.81 \times V_R \text{ (J4 Page 361)}$$

$$V_R = 67 - 41R^{(-0.1)} = 46.45 \text{ m/s} \approx 47 \text{ m/s}$$

Design Wind Pressure:

$$P_w = 0.5 \times \rho_{air} \times (V_{des}^2) \times C_{fg} \times C_{dyn} = 0.5 \times 1.2 \times 1474.5 \times 1 \times 1.2 = \mathbf{1061.6 \text{ Pa}}$$

Force (F):

$$F = P_w \times G \times C_f \times A_f = \mathbf{163.132 \text{ kN}}$$

Moment at the base of the skirt:

$$M = F \times h_2 = 163.132 \times 35.3 = \mathbf{5758.6 \text{ kNm}}$$

Seismic Load AS 1170.4

- Annual P of exceedance (Importance 4): 1/800
- Acceleration coefficient of Geelong = 0.10
- Site factor for general structures: $S = 1.5 / 1$ (Page 29)
- **Structure Classification:** Type II (Covers 4 Story Building)
- Importance level = 1.00

Earthquake Design Category:

$$a_s = 0.10 \times S = 0.15$$

Category is C if $0.1 \leq a_s \leq 0.2 \rightarrow H2$ (Sections 6, 7; Clause 4.4 (4.3.2, 4.3.3))

Height Amplification Factor a_x :

$$a_x = 1 + (h_x / h_a) = 1 + (30 + 1.1) / (30 + 1.1) = \mathbf{2.135 \text{ m}}$$

Earthquake Base Shear

- $V = I * (C_s / R_f) * G_g$ ($V \geq 0.01 G_g$)
- G_g , Gravity load = operating vessel weight = $327,760 \text{ kg} \times g$
- I : Importance factor = 1.00
- Earthquake design coefficient, $C = 1.25a / T^{(2/3)} = 1.25 \times 0.1 / T^{(2/3)} = 0.15$
- S : Site factor = 1.5
- Structural response factor, $R_f = 2.1$
- Fundamental Period: $T = h_n / 46 = 35.3 / 46 = 0.7674 \text{ s}$

Range:

$$I (2.5a / R_f) G_g \geq V \geq 0.01 G_g \quad (\text{Page 38})$$

$$\text{So, } V = 1 \times (0.15 \times 1.5 / 2.1) \times 3231 \text{ kN} = \mathbf{346.6 \text{ kN}}$$

$$\mathbf{385.2 \geq 346.6 \geq 387.35 \text{ kN}}$$

V is the total horizontal base shear force.

Calculation From Project Guide:

- $V = (ZIC / R_w) \times W$
- $T = (C_t \times H^{3/4}) = 1.234 \text{ s}$
- $C = 2.35 \times (S / T^{2/3}) = 2.04$
- $V = 51131 \text{ lbf} = 227.44 \text{ kN (shear)}$
- $F_t = 0.07 \times TV = 4416.7$

Moment:

$$M = (F_x H) + ((V - F_L) / 2H) \times 2H/3$$

$$M = 4416.7 \times (11.581) + (51.13 - 4416.7) \times (2 \times 11.581) / 3$$

$$M = 911,815.3 \text{ lb-ft} \Rightarrow 5583.47 \text{ kNm}$$

Max moment is produced by wind load at base.

Thus, max moment is from now on will be by wind force.

Thin Shell Ellipsoid

Types of Loadings:

- Internal pressure (P_i)
- External pressure (P_w) (Wind load)
- Self-weight

Assumptions made: Comparing Internal pressure, Wind pressure and pressure due to mass of the fluid is negligible.

Stresses at the equator:

$$\sigma_l = \frac{Pa}{2t} ,$$

$$\sigma_l = \frac{(1.71) \cdot 4270}{4 \cdot 25} = 73 \text{ MPa}$$

$$\sigma_h = \frac{Pa}{t} \left(1 - \frac{a^2}{2b^2} \right)$$

$$\sigma_h = \frac{(1.71) \cdot 4270}{2 \cdot 25} \left(1 - \frac{2135^2}{2 \cdot 1100^2} \right) = -129 \text{ MPa}$$

P (working Pressure)	1.71Mpa
t	25mm
a	2135mm
b	1100mm

Known Parameters	
Mean Diameter (D_m):	4.27 m
Skirt Thickness (t):	0.013 m
Design Pressure (P_d):	1.946 MPa
Wind Force:	199,000 N
Skirt Height (h):	4.2 m
Weight of Vessel (Operating):	329,700 kg
Weight of Vessel (Empty):	98,900 kg

Stress Calculations for Vertical Pressure Vessel Shell and Skirt (AS 1210-2010)

Shell: Wind Load, Internal Pressure, and Operating Weight (Sub section 3.7.5)

Bending Stress (S_b):

Bending Moment due to Wind Load:

$$M = F * h = 199,000 * 15 = 2,985,000 \text{ Nm (at midpoint)}$$

$$S_b = (4 * M) / (\pi * D_m^2 * t)$$

$$S_b = (4 * 7,024,700) / (\pi * (4.27)^2 * 0.025) = -8.38 \text{ MPa}$$

Circumferential Stress (S_c):

$$S_c = (P * D_m) / (2 * t)$$

$$S_c = (1.71 * 10^6 * 4.27) / (2 * 0.025) = 146 \text{ MPa}$$

Weight Stress (S_w):

The weight stress is calculated as:

$$S_w = W / (\pi * D_m * t)$$

$$S_w = (3.23 * 10^6) / (\pi * 4.27 * 0.025) = -9.65 \text{ MPa}$$

Combined Stress (S_E):

$$S_e = \max\{S_c, |S_c/2 + S_w + S_b|, S_c/2 + S_w S_b, S_w + S_s/2, S_s - S_c/2\}$$

$$S_E = 146 \text{ MPa}$$

Skirt: Wind Load and Operating Weight (Sub section 3.7.5)

Bending Moment due to Wind Load:

$$M = F * h = 199,000 * 35.3 = 7024,700 \text{ Nm (max value)}$$

Bending Stress (S_b):

$$S_b = (4 * M) / (\pi * D_m^2 * t)$$

$$S_b = (4 * 7,024,700) / (\pi * (4.27)^2 * 0.013) = 37.73 \text{ MPa}$$

Circumferential Stress (S_c):

$$S_c = (P * D_m) / (2 * t)$$

$$S_c = (0 * 10^6 * 4.27) / (2 * 0.013) = 0$$

Weight Stress (S_w):

The weight stress is calculated as:

$$W = 329,700 \cdot 9.81 = 3.23 \cdot 10^6 \text{ N}$$

$$S_w = W / (\pi \cdot D_m \cdot t)$$

$$S_w = (3.23 \cdot 10^6) / (\pi \cdot 4.27 \cdot 0.013) = \mathbf{18.54 \text{ Mpa}}$$

Combined Stress (S_E):

$$S_e = \max\{S_c, |S_c/2 + S_w + S_b|, S_c/2 + S_w S_b, S_w + S_s/2, S_s - S_c/2\}$$

$$S_E = \max(0, 19.18, 56, 19, 56) = \mathbf{56 \text{ MPa}}$$

Stress Calculations for Vertical Pressure Vessel Ellipsoid bottom cap.

$$\sigma_l = \frac{PR}{2t \cos \theta_l} + \frac{W}{2\pi R t \cos \theta_l}$$

At bottom,

$$\cos \theta_l = 1, W = 0$$

$$\sigma_l = \frac{(1.71) \cdot 0.11}{2 \cdot 0.025} = \mathbf{37.62 \text{ MPa}}$$

$$\frac{\sigma_l}{r_l} + \frac{\sigma_h}{r_h} = \frac{P}{t}$$

$$\sigma_h = r_h \cdot \left(\frac{P}{t} - \frac{\sigma_l}{r_l} \right)$$

$$\sigma_h = 1100 \cdot \left(\frac{1.71}{25} - \frac{37.62}{1100} \right) = \mathbf{37.62 \text{ MPa}}$$

Ellipsoid head (at equator)	
σ_l	73 MPa
σ_h	-129 MPa
Shell	
σ_l	73 MPa
σ_h	146 MPa
Ellipsoid Bottom	
σ_l	37.62 MPa
σ_h	37.62 MPa
Skirt	
σ_l	56 MPa
σ_h	0

Table 2: Stress with LPG

c. Comparison between the hoop and longitudinal stress

At the top of the pressure vessel hoop and longitudinal stress is equal, there is no stress induced by weight of the vessel and the stress from wind loading is negligible compared to internal pressure.

- Ellipsoidal **Head (Equator)**: **Hoop stress is higher** (compressive) due to the curved geometry.
- Cylindrical **Shell**: **Hoop stress is higher** (about twice as large as longitudinal stress), typical for cylindrical sections under internal pressure.
- Ellipsoidal **Bottom**: **Hoop and longitudinal stresses are equal** due to the balanced stress distribution in the bottom part of the head. Weight is carried by the skirt, thus no effect of fluid mass. Also, the effect of wind load is negligible
- Skirt: **Only longitudinal stress exists** due to the weight of the vessel, and no hoop stress is present as the skirt does not contain internal pressure.

d. Empty and hydrostatic condition analysis

Empty vessel

Ellipsoid head:

Only pressure due to wind is acting = 1297 Pa (calculated earlier)

Buckling under External Pressure:

Thin-walled vessels are prone to **buckling** under external pressure. The critical buckling pressure P_{cr} can be determined using the following equation for a cylindrical vessel:

$$P_{cr} = \frac{2 \cdot E}{1 - \nu^2} \cdot \left(\frac{t}{r} \right)^3$$

$P_{cr} = 0.74 \text{ MPa}$ (safe)

Skirt: Wind Load and Operating Weight (Sub section 3.7.5)

Bending Moment due to Wind Load:

- $M = F \cdot h = 199,000 \cdot 35.3 = 7024,700 \text{ Nm}$ (max value)

Bending Stress (S_b):

- $S_b = (4 \cdot M) / (\pi \cdot D_m^2 \cdot t)$
 $S_b = (4 \cdot 7,024,700) / (\pi \cdot (4.27)^2 \cdot 0.013) = \mathbf{37.73 \text{ MPa}}$

Circumferential Stress (S_c):

$$S_c = (P * D_m) / (2 * t)$$

$$S_c = (0 * 10^6 * 4.27) / (2 * 0.013) = 0$$

Weight Stress (S_w):

$$W = 98,900 * 9.81 = 0.97 * 10^6 \text{ N}$$

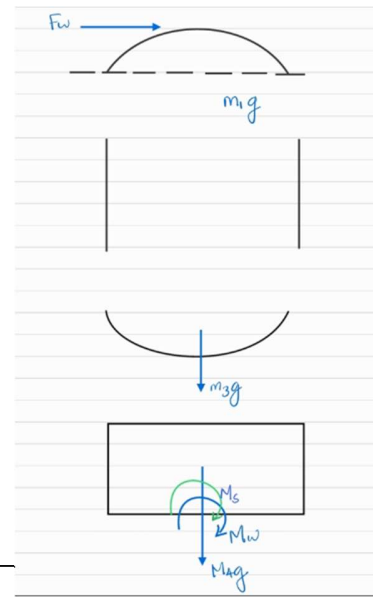
$$S_w = W / (\pi * D_m * t)$$

$$S_w = (3.23 * 10^6) / (\pi * 4.27 * 0.013) = \mathbf{5.56 \text{ MPa}}$$

Combined Stress (S_E):

$$S_e = \max\{S_c, |S_c/2 + S_w + S_b|, S_c/2 + S_w S_b, S_w + S_s/2, S_s -$$

$$S_E = \mathbf{43.3 \text{ MPa}}$$

**Shell Wind Load and Operating Weight****Bending Moment due to Wind Load:**

- $M = F * h = 199,000 * 15 = 2985000 \text{ Nm}$ (moment at the middle of vessel)

Bending Stress (S_b):

$$S_b = (4 * M) / (\pi * D_m^2 * t)$$

$$S_b = (4 * 7,024,700) / (\pi * (4.27)^2 * 0.013) \\ = \mathbf{8.33 \text{ MPa}}$$

Circumferential Stress (S_c):

$$S_c = (P * D_m) / (2 * t)$$

$$S_c = (0 * 10^6 * 4.27) / (2 * 0.013) = 0$$

Weight Stress (S_w):

$$W = 60,900 * 9.81 = 0.597 * 10^6 \text{ N}$$

$$S_w = W / (\pi * D_m * t)$$

$$S_w = (3.23 * 10^6) / (\pi * 4.27 * 0.013) = \mathbf{1.8 \text{ MPa}}$$

Ellipsoid head	
σ_l	0
σ_h	0
Shell (middle)	
σ_l	10.2
σ_h	0
Ellipsoid Bottom	
σ_l	0
σ_h	0
Skirt	
σ_l	43.3
σ_h	0

Table 3: Stress with empty vessel

Combined Stress (S_E):

$$S_e = \max\{S_c, |S_c/2 + S_w + S_b|, S_c/2 + S_w S_b, S_w + S_s/2, S_s - S_c/2\}$$

$$S_E = \mathbf{10.2 \text{ MPa}}$$

Hydrostatic condition

At ellipsoid top equator: Internal pressure due to water is $= h \cdot \rho \cdot g = 1.1 \cdot 1000 \cdot 9.81 = 0.01 \text{ MPa}$

That's negligible. So we can assume the stress at ellipsoid head is zero.

At shell :

Internal pressure avg: $1000 \cdot 10 \cdot 30 \cdot 0.5 = 0.15 \text{ MPa}$

$$\text{Hoops stress } \sigma_h = \frac{PR}{t} =$$

$$0.15 \cdot 2.135 / 0.025 = 12.81 \text{ MPa}$$

$$\text{Longitudinal Stress, } \sigma_l = \frac{PR}{2t} = 6.405 \text{ MPa}$$

At skirt:

No internal pressure so, hoop's stress is 0;

And longitudinal stress is

$$S_e = \max\{S_c, |S_c/2 + S_w + S_b|, S_c/2 + S_w S_b, S_w + S_s/2, S_s - S_c/2\}$$

$$S_E = 64.23 \text{ MPa}$$

Ellipsoid bottom

$$\sigma_l = \frac{(0.3) \cdot 0.11}{2 \cdot 0.025} = \mathbf{0.66 \text{ MPa}}$$

$$\frac{\sigma_l}{r_l} + \frac{\sigma_h}{r_h} = \frac{P}{t}$$

$$\sigma_h = r_h \cdot \left(\frac{P}{t} - \frac{\sigma_l}{r_l} \right)$$

$$\sigma_h = 1100 \cdot \left(\frac{1.71}{25} - \frac{37.62}{1100} \right) = \mathbf{0.66 \text{ MPa}}$$

Ellipsoid head	
σ_l	0
σ_h	0
Shell	
σ_l	6.405 MPa
σ_h	12.81 MPa
Ellipsoid Bottom	
σ_l	0.66 MPa
σ_h	0.66 MPa
Skirt	
σ_l	64 MPa
σ_h	0

e. Minimum Acceptable Thickness of skirt

Using standards: From AS 1210-2010 Section 3.7.5,

$$S_{CL} = 0.605 \cdot 200 \cdot 10^3 \cdot \frac{x}{4270} \cdot \frac{\left(2880 + \frac{4270}{x}\right)}{\left(3200 + 10 \cdot \frac{4270}{x}\right)}$$

$$S_{CL} = \frac{P_e \cdot D_m}{2t}, \quad A_a = 1.1 \left(\frac{t}{D_m}\right)^2$$

$$P_e = \frac{2EA_a t}{D_m} \quad \Rightarrow \quad P_e = \frac{2E \cdot 1.1 \left(\frac{t}{D_m}\right)^2 t}{D_m}$$

Thickness $t = 4.1\text{mm}$

Using equations:

Stress at the Skirt Bottom:

- $N = mg$ (The weight of the vessel and its contents acts on the skirt)
- Stress due to bending Moment

Longitudinal Stress:

$$\sigma_l = \frac{(mg)}{2\pi r t} + \frac{4M}{\pi * D^2 * t} \sim = 155 \text{ MPa}$$

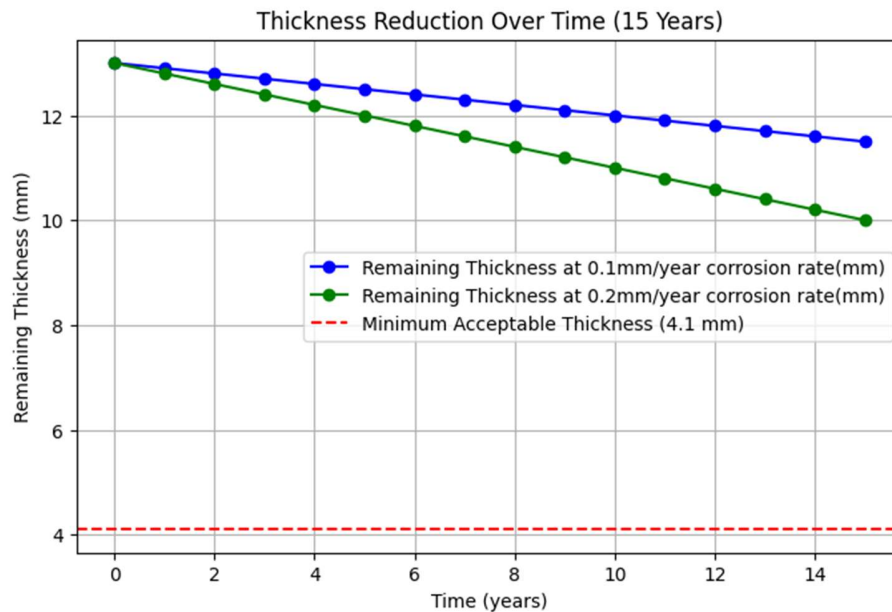
Thickness $t \geq 3.7\text{mm}$

Assumptions:

- Corrosion Rate: Assume a uniform corrosion rate of 0.1 mm/year. This rate can be based on industry standards or site-specific data (adjustable based on actual environmental conditions or historical data).
- Operating Conditions: Assume no significant change in operating conditions over time.

The thickness reduction over time is linear if the corrosion rate remains constant. The remaining thickness at any time t in years can be calculated as:

$$T_{\text{remaining}} = T_{\text{initial}} - 0.1 \cdot \text{year}$$



f. Minimum acceptable thickness of the shell, top and bottom heads analysis.

<i>Parameter</i>	<i>Description</i>	<i>Unit</i>
P	Design pressure or maximum allowable working pressure	psi
S	Stress value of material	psi
E or η	Joint efficiency (=1)	-
R	Inside radius	inches
D	Inside diameter	inches
t	Wall thickness	inches

Assumptions:

The pressure vessel is only subjected to internal pressure.

Shell:

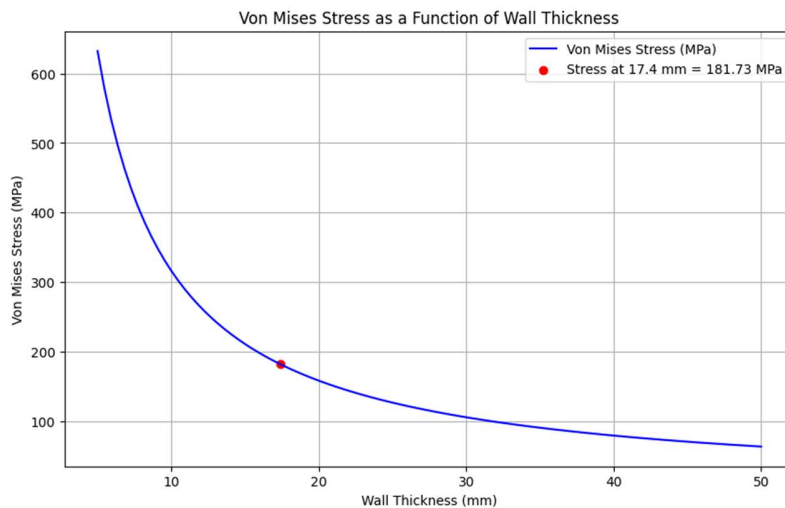
$$t = \frac{PR}{SE - 0.6P}$$

$$t = \frac{248.0145 * 84.05512}{30240.368 * 1 - 0.6 * 248.0145} = 17.596612 \text{ mm}$$

Ellipsoid head

$$t = \frac{PD}{2SE - 0.2P}$$

$$t = \frac{248.0145 * 2 * 84.05512}{2 * 30501.436 * 1 - 0.2 * 248.0145} = 17.374 \text{ mm}$$



The **minimum thickness point** on the graph is the point where the material thickness is just enough to withstand the internal pressure without failing. Physically, it represents the **critical thickness** needed for safe operation of the pressure vessel. Any thickness below this value would lead to structural failure, while increasing the thickness reduces stress but increase material usage and cost.

g. Vibration Analysis

Actual Period of Vibration $T = 0.0000265 \left(\frac{H}{D}\right)^2 \sqrt{\frac{wD}{t}}$,

$T = 1.753 \text{ s}$

Maximum allowable period of Vibration $T_a = 0.80 \sqrt{\frac{WH}{V_g}}$

- D = Outside diameter of vessel, ft.
- H = Length of vessel including skirt, ft.
- g = 32.2 ft. per sec. squared, acceleration
- t = Thickness of skirt at the base, in.
- V = Total shear, lb. CW ; see page 61
- W = Weight of tower, lb.
- w = Weight of tower per foot of height, lb.

$T_a = 5.72s$

- **Actual Period of Vibration (T):** 1.753s
- **Maximum Allowable Period of Vibration (T_a):** 5.72s

Analysis:

Since T (1.753s) is less than T_a (5.72s), the actual period of vibration is within the acceptable range. This indicates that the pressure vessel is likely to be safe from excessive vibrations.

Safety Judgment:

Based on the vibration analysis, the pressure vessel is **judged to be safe** from a vibration perspective. The actual period of vibration is significantly lower than the maximum allowable period, suggesting that the vessel is unlikely to experience excessive vibrations that could compromise its integrity or safety.

h. Role of Stress-Relieving

The absence of stress-relieving during manufacturing of a pressure vessel can significantly impact the integrity and longevity of the vessel. Stress-relieving is a heat treatment process that reduces residual stresses in materials, which develop during welding or forming. Without this treatment, the pressure vessel is more prone to failure due to stress concentrations at welds, leading to reduced fatigue life and a higher risk of cracking under cyclic loading or thermal stresses.

Confirmation methods:

NDT, checking manufacturing records could confirm this.

Changes to Stress Analysis:

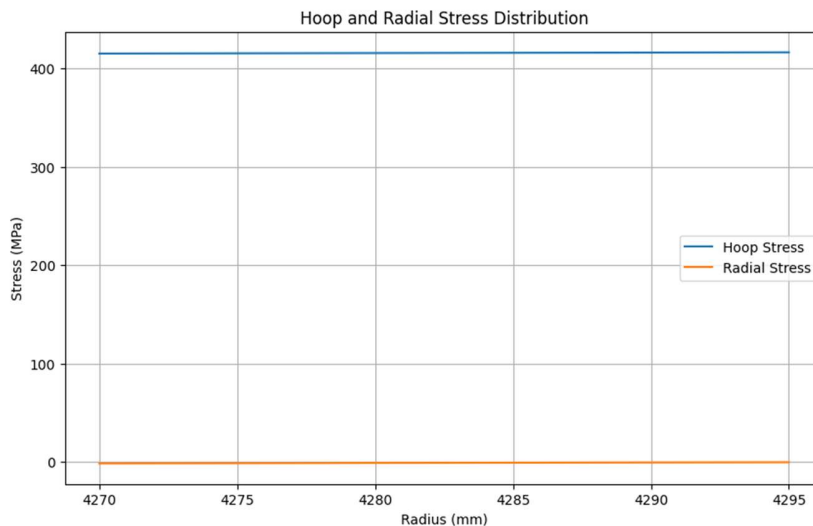
Without stress-relieving, Increase the Stress Allowance: Incorporate a higher safety factor to account for the unknown distribution of residual stresses. Analyze Welded Joints More Critically: Focus on areas prone to higher residual stress, such as welds and sharp corners. Conduct a Fatigue Analysis: Include fatigue life estimation under cyclic loading conditions.

For a thick wall stress at any point r, is given by

$$\sigma_{\theta\theta} = S_{yld} + \sigma_{rr} = S_{yld} \left(1 + \log_e \left(\frac{r}{a} \right) \right) - p_i$$

$$\sigma_{rr} = S_{yld} \log_e \left(\frac{r}{a} \right) - P_i$$

At yield the internal pressure is much higher than the operational pressure, thus it is okay to operate the vessel without stress relief treatment.



i. Thermal Stress

Assumption:

material is steel with coefficient of expansion $\alpha = 1.9 \times 10^{-6} / ^\circ\text{C}$

Maximum thermal fluctuation possible is 60 degrees (Geelong doesn't have variation more than 60 degrees)

$$\sigma_l = \frac{PR}{2t}, \quad \sigma_h = \frac{PR}{t} \quad \text{and strain } \epsilon_h = \frac{1}{E}(\sigma_h - \nu\sigma_e)$$

$$\text{So } \epsilon_h = \frac{1}{200500}(146 - 0.3 * 73) = 0.6\text{mm}$$

$$u_{\Delta T} = \alpha r \Delta T = r^* \epsilon_H = 1.9 \times 10^{-6} * 2.1475 * 60 = 0.245\text{mm}$$

so thermal stress will introduce an additional strain of 0.245mm to the cylinder but this is not the case in skirt and shell joint, there it is constrained. **Thermal expansion** can cause stress concentrations, particularly near welded joints or areas of differing material thickness. Over time, this can lead to **thermal fatigue**, crack initiation, or even failure if not properly accounted for. To manage this, stress-relief methods, material selection, and thermal insulation should be considered in the vessel design to mitigate the impact of thermal stress on its integrity

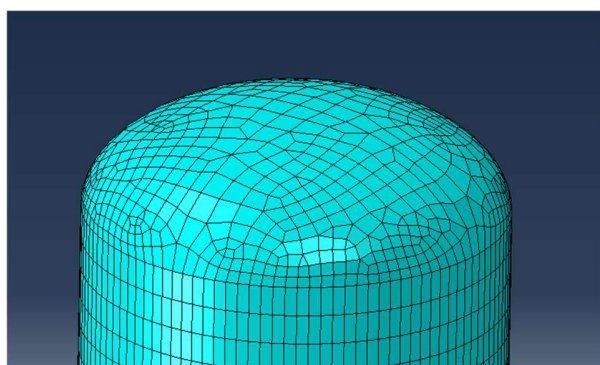
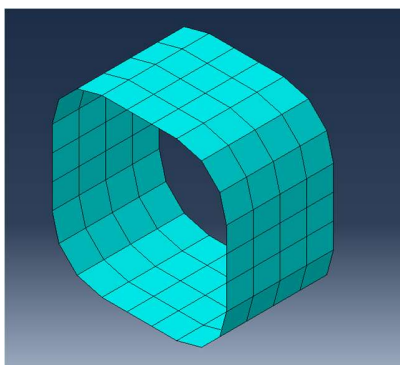
j. Initial thickness variation analysis

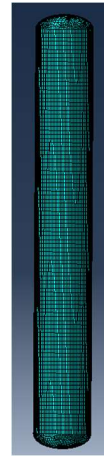
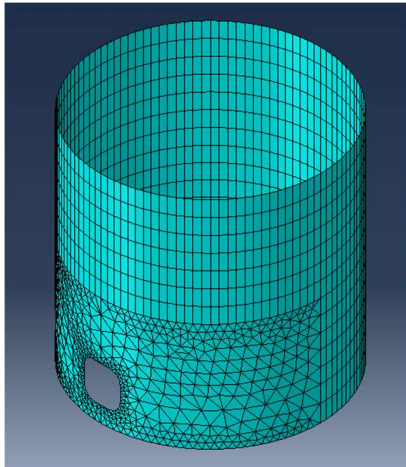
Assuming the initial thickness of 12mm and corrosion rate of 0.2mm/year, it will take 35 years to reach 5mm thickness, as per the analysis the minimum thickness is in the range of 3.5 to 4.5mm. But there are regions where the stress is concentrated. Also, vessel will undergo thermal cycle during this period. This could be analyzed by fatigue analysis. From the initial assumption of 13mm (conservative thickness) changing to 12mm will not significantly change the values. 13mm assumption is based on the advice from a senior mechanical engineer and suggested not to consider the minimum thickness obtained by laser scan. Moreover, recommending a treatment process where the thickness is below 13mm is recommended instead of using the minimum thickness of 3.56mm. Modelling Analysis Part 1

Meshing and Element Type Selection

The pressure vessel was modelled as a shell, as the thickness of the parts of the vessel are 10 times less than the radius, which limits the meshing elements to quad, quad-dominated and tri. For the shell and the back of skirt, a structured quad mesh was used which is the most accurate available mesh. A quad-dominated mesh was used on the heads due to the complexity of the geometry. A tri mesh was used for the area around the manhole in the skirt. This is the least accurate mesh; however, it was used to prevent distorted elements from complex geometry.

An encastre boundary condition at the bottom of the skirt and contact ties between the skirt and the shell, as well as the skirt and the manhole shroud, with these welded joints perfectly transfer force.





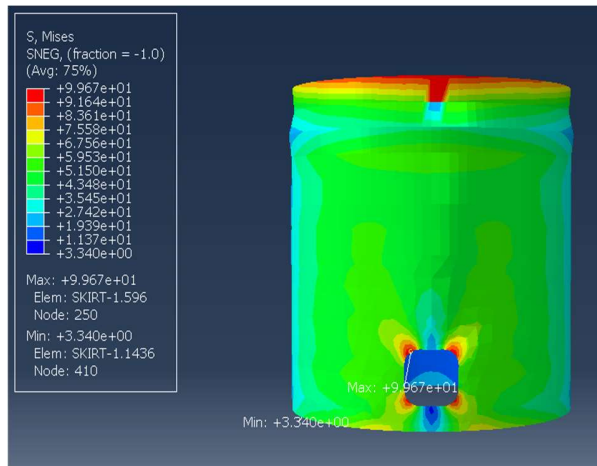
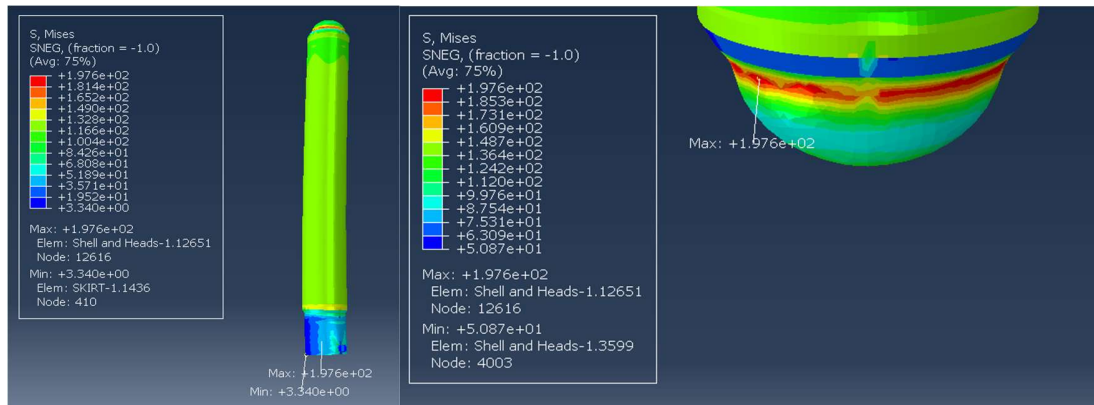
a. Wind/Seismic Load Analysis

Loads:

- Wind Load - 199kN was applied at the very top of the vessel (on the head) pointing tangentially towards the manhole cover, in order to simulate the worst-case scenario.
- Gravity - The density of the vessel was calculated assuming constant density of 8190 kg/m^3 with a gravitational acceleration of 9.8 m/s^2
- Weight of Fluid – A Body Force for the weight of the fluid was applied to the bottom head of $4.78 \times 10^{-6} \text{ N/mm}^3$ assuming the tank is full of LPG.
- Hydrostatic Pressure - applied to the shell and heads with the pressure at the bottom of the tank being 0.1527 MPa (ignoring variation due to the height of the ellipsoidal heads) and zero at the top.
- Operating Pressure – Assumed the Operating Pressure referred to the max pressure in the tank so an additional pressure of 1.555 MPa applied uniformly in the heads and shell

Stresses:

- Maximum Stress Located at the interface between the shell and the bottom head with a von Mises Stress of 197 MPa
- The Maximum Stress in the Skirt is located above the corner of the hole from the manhole with a von Mises Stress of 99.72 MPa
- The Maximum Stress of the Shell is 134.2 MPa



Ellipsoid head (at equator)	
σ_{vm}	177.16 MPa
Shell	
σ_{vm}	126.43 MPa
Ellipsoid Bottom	
σ_{vm}	37.62 MPa
Skirt	
σ_{vm}	56 MPa

To compare the analytical and numerical results, we have to convert the hoop and longitudinal forces into von Mises stresses using the following formula:

$$\sigma_{VM} = \sqrt{\sigma_h^2 - \sigma_h \sigma_l + \sigma_l^2}$$

The maximum stress occurs in the same location which a similar magnitude which the numerical maximum stress being 197.6 MPa and the analytical maximum stress being 177.16 MPa. These results are expected to be different as the numerical model has a more accurate representation of the mass distribution, as well as modelling the hydrostatic force acting on the vessel. Additionally, the numerical method was limited by the number of mesh elements, which means there is some inherent error with the solution.

The analytical solution failed to identify the stress concentrations at the manhole, which is a significant location of stress. It was neglected due to the complexity of the geometry.

Both solutions demonstrate that the maximum stresses acting on the vessel are below the Yield Stresses of the material, which demonstrates why the vessel should remain standing for the moment

b. Safety, Lifetime, and Recommendations

The Safety Factor for the head is given by:

$$FoS = \frac{S_y}{\sigma_{max}} = \frac{210}{197.6} = 1.064$$

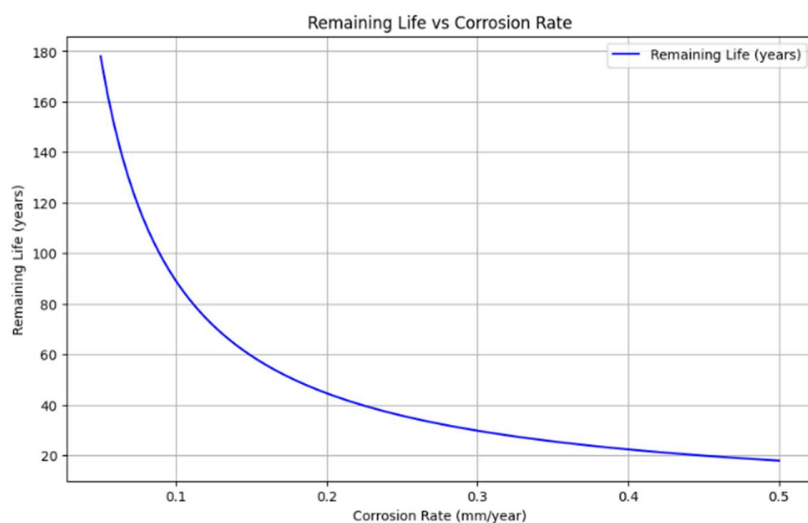
For a highly dangerous and important vessel, this safety standard is not high enough to be considered safe. Additionally, the FoS for the skirt and shell are 1.554 and 1.555 respectively which are far more reasonable, however, it is recommended that the safety factor of pressure vessel be at least 2.

Lifetime Prediction

The remaining operational lifetime of the vessel can be predicted based on the **current condition of the vessel skirt thickness** and the **corrosion rate**. Given the current thickness of the skirt in range of 3.56 to 17 mm, 13mm could be taken as a weighted average and a **corrosion rate of 0.1 mm per year**, we can calculate the **remaining operational lifetime** before the vessel reaches the **minimum acceptable thickness of 4.1 mm**.

$$\text{Remaining Lifetime} = \frac{\text{Current Thickness} - \text{Minimum Thickness}}{\text{Corrosion Rate}}$$

$$\text{Remaining Lifetime} = \frac{13 - 4.1}{0.1} = 89 \text{ years}$$



This prediction assumes steady corrosion without any significant operational changes. If additional inspections reveal faster corrosion rates or other unforeseen issues, the lifetime estimate would need to be revised. It is crucial to maintain regular inspections to verify this rate and update the prediction accordingly. Also assumed initial

thickness of 13mm may not be the best suitable thickness, which could further decrease the remaining life of the vessel.

Practical Recommendations

To ensure the safety and extended operational life of the vessel, the following practical and constructive recommendations are made:

1. **Increase Inspection Frequency:** Regular inspections of the vessel are essential, particularly focused on areas prone to corrosion, such as the skirt and welded joints. Ultrasonic thickness measurements and non-destructive testing (NDT) should be conducted annually to monitor the rate of degradation and assess any new defects.
2. **Corrosion Mitigation:** Applying a corrosion-resistant coating to the internal and external surfaces of the skirt can slow down the rate of material loss. Cathodic protection should also be considered to mitigate corrosion, especially in areas exposed to harsh environmental conditions.
3. **Fatigue and Stress Analysis:** Since the vessel operates under cyclical loading (pressure fluctuations), it is important to perform a detailed fatigue analysis. This will help determine how cyclic stresses might reduce the vessel's operational life and highlight areas that could require strengthening or retrofitting.
4. **Retrofitting and Repairs:** If significant thickness loss is observed, management should consider retrofitting or strengthening the skirt with additional material or supports to increase its safety margin. In critical regions (like the skirt or welds), post-weld heat treatment (PWHT) may reduce residual stresses and increase the vessel's fatigue life.
5. **Monitor Operational Conditions:** Keep detailed records of operational conditions, such as pressure, temperature fluctuations, and loading cycles, as any significant deviations could accelerate wear or reduce the vessel's safety. Immediate corrective actions should be taken if operational parameters exceed design limits.

Part 2: Analytical Study

Outer Radius of Shell and Spherical Head	$R = b = 1.025 \text{ m}$
Thickness of Vessel	$t = 0.02 \text{ m}$
Yield Strength	$S_y = 620 \times 10^6 \text{ Pa}$
Inner Radius	$a = R - t = 1.005 \text{ m}$
Radius of plastic-elastic interface	$c = R - t + 0.005 = 1.01 \text{ m}$

a. Yield Pressure for Thin-Walled Vessel

A thin-walled formula is valid when the thickness must be 10 times smaller than the radius: $\frac{R}{t} = 51.25 > 10$

It is assumed the vessel is static and the weight of the fluid within the vessel is negligible. The stresses in the cylinder and shell respectively are:

$$\sigma_{h,cyl} = \frac{pR}{t}, \sigma_{l,cyl} = \frac{pR}{2t}, \sigma_{h,sph} = \frac{pR}{2t}, \sigma_{l,sph} = \frac{pR}{2t}$$

Therefore, the maximum stress occurs on the cylindrical part of the shell meaning the yield pressure is given by:

$$p_y = \frac{S_y t}{R} = \frac{620 \times 10^6 \times 0.02}{1.025} = 12.10 \text{ MPa}$$

b. Yield Pressure for Non-Thin-Walled-Vessel

For a thick-walled analysis, it is assumed, as shown above, that the yield point acts on the cylindrical part of the shell. It is also assumed that the external pressure is negligible when compared with the internal pressure required to yield the steel.

The hoop and longitudinal stress respectively are:

$$\sigma_{rr} = \frac{a^2 p_y}{b^2 - a^2} \left(1 - \frac{b^2}{r^2} \right), \sigma_{\theta\theta} = \frac{a^2 p_y}{b^2 - a^2} \left(1 + \frac{b^2}{r^2} \right)$$

Applying Tresco's Failure Criteria:

$$S_y = \sigma_{hh} - \sigma_{\theta\theta} = \frac{2(R-t)^2 R^2 p_y}{R^2 - (R-t)^2} \times \frac{1}{r^2}$$

Rearranging gives:

$$p_y = \frac{S_y \times r^2 \times (R^2 - (R-t)^2)}{2(R-t)^2 R^2}$$

Therefore, the minimum yield pressure occurs at the minimum possible radius ($r = R - t$) meaning:

$$p_y = S_y \frac{t}{R} \left(1 - \frac{t}{2R}\right) = 11.98 \text{ MPa}$$

Completing a comparative analysis, the approximated thin-wall pressure is within 10% of the thick-walled value. The value of the thin-walled yield pressure is greater than the thick-walled yield pressure, which is to be expected, as the inner radius is smaller compared to the thin-walled vessel.

c. Pressure to Yield Vessel to 5mm

To calculate the yield pressure, it was assumed that the material behaved perfectly elastic after the yield point. The yield pressure can be calculated from the following formula assuming the Tresca Failure Criteria is valid:

$$r \frac{d\sigma_{rr}}{dr} = \sigma_{\theta\theta} - \sigma_{rr} = S_y$$

Integrating and substituting boundary conditions $\sigma_{rr} = -p_i$ when $r = a$, and substituting in hoop and longitudinal stress for $r = c$ and $a = c$ using equation used in previous question:

$$p_y = S_y \left(\ln\left(\frac{c}{a}\right) + \frac{1}{2} \left(1 - \frac{c^2}{b^2}\right) \right) = 12.084 \text{ MPa}$$

d. Residual Stress Determination

The plastic region is defined when $a < r < c$. Assuming the internal pressure is perfectly and uniformly counteracted by the radial stress:

$$\sigma_r = S_y \ln\left(\frac{r}{a}\right) - p_i, \quad \sigma_\theta = S_y \left(\ln\left(\frac{r}{a}\right) + 1\right) - p_i$$

where $p_i = 12.084 \text{ MPa}$

The elastic pressure is given by the radial stress at $r = c$:

$$p_e = -S_y \ln\left(\frac{c}{a}\right) + p_i = 9.00 \text{ MPa}$$

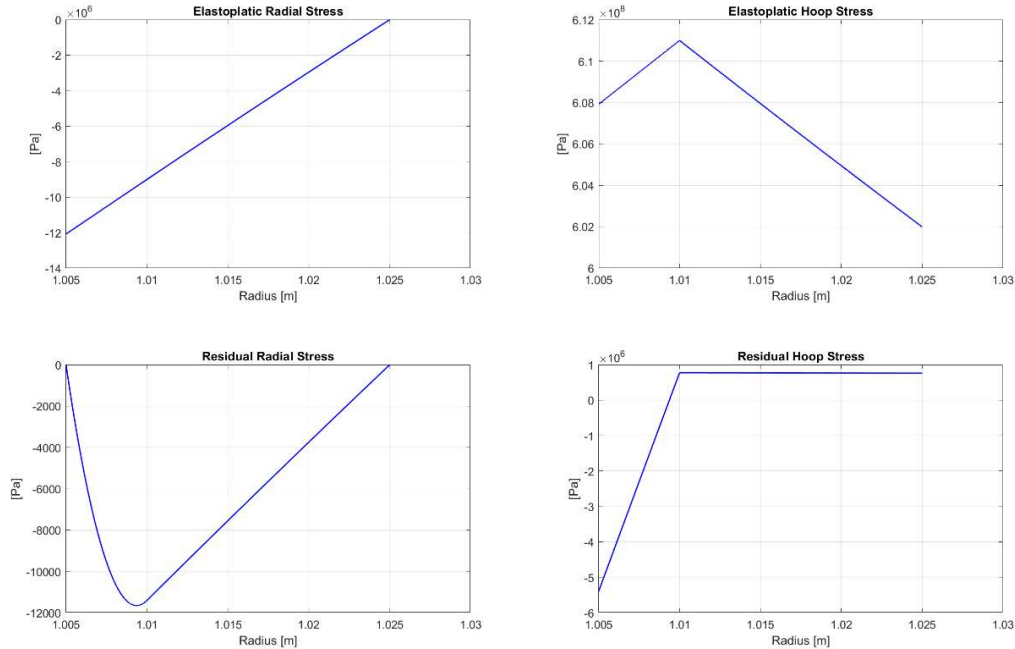
The elastic region is defined when $c < r < b$. The stresses are given by:

$$\sigma_{rr} = \frac{c^2 p_e}{b^2 - c^2} \left(1 - \frac{b^2}{r^2}\right), \quad \sigma_{\theta\theta} = \frac{c^2 p_e}{b^2 - c^2} \left(1 + \frac{b^2}{r^2}\right)$$

The purely elastic stress is defined when $a < r < c$ and given by:

$$\sigma_{rr} = \frac{a^2 p_i}{b^2 - a^2} \left(1 - \frac{b^2}{r^2}\right), \quad \sigma_{\theta\theta} = \frac{a^2 p_i}{b^2 - a^2} \left(1 + \frac{b^2}{r^2}\right)$$

Therefore, the residual stress can be taken as the difference between the purely elastic stresses with the elastoplastic stresses shown in the following plots



e. Hoop Stress Calculation on Venus

The pressure vessel on Venus is subject to atmospheric pressure of 92 bar ([Venus Fact Sheet \(nasa.gov\)](https://www.nasa.gov/factsheets/venusfact.html)) with the pressure having been equalised from inside and outside of the vessel.

Therefore, the hoop stress of the cylinder is given by:

$$\sigma_{\theta\theta} = \frac{(a^2 - b^2)p_{\text{venus}}}{b^2 - a^2} = -p_{\text{venus}} = -9.2 \text{ MPa}$$

For the spherical caps, the hoop stress is given by:

$$\sigma_{\theta\theta} = \frac{p_{\text{venus}}b^3}{2r^3} \times \frac{2r^3 + a^3}{a^3 - b^3} - \frac{p_{\text{venus}}a^3}{2r^3} \times \frac{2r^3 + b^3}{a^3 - b^3} = -p_{\text{venus}} = -9.2 \text{ MPa}$$

The average temperature on Venus is 464 °C which, with some metals, can reduce the yield strength to 25% ([HYPERLINK "https://www.engineeringtoolbox.com/metal-temperature-strength-d_1353.html"](https://www.engineeringtoolbox.com/metal-temperature-strength-d_1353.html) https://www.engineeringtoolbox.com/metal-temperature-strength-d_1353.html)

$$S_{y,\text{venus}} = 0.25S_y = 155 \text{ MPa}$$

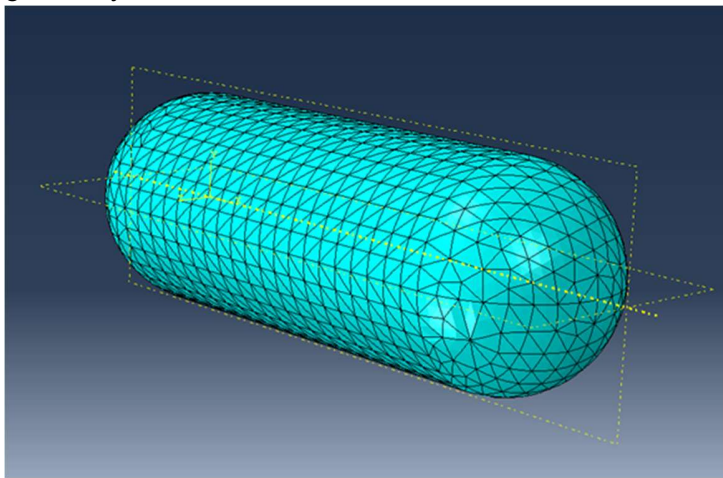
Assuming no chemical reactions or unforeseen loading, the hoop stress in the vessel is equal to

$$-9.2 \text{ MPa} .$$

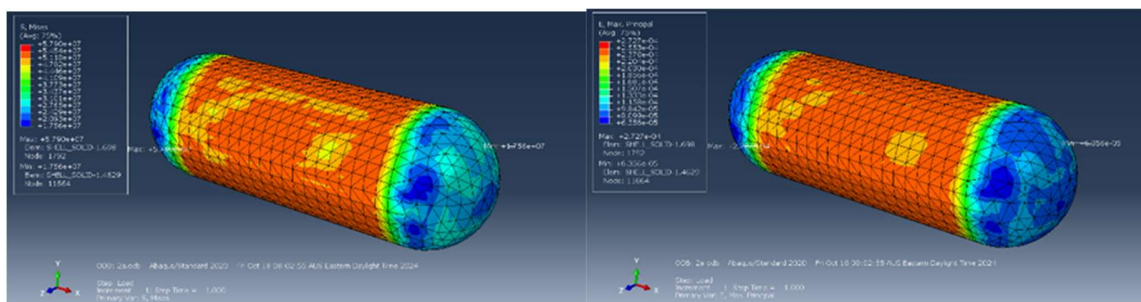
Part 2: Numerical Study

a. Meshing and Stress Analysis with Tetrahedral Elements

In this simulation, a tetrahedral mesh was utilized after longitudinal partitioning in the x-y and x-z planes as shown. The selection of this element enables the meshing of the highly curved geometry.



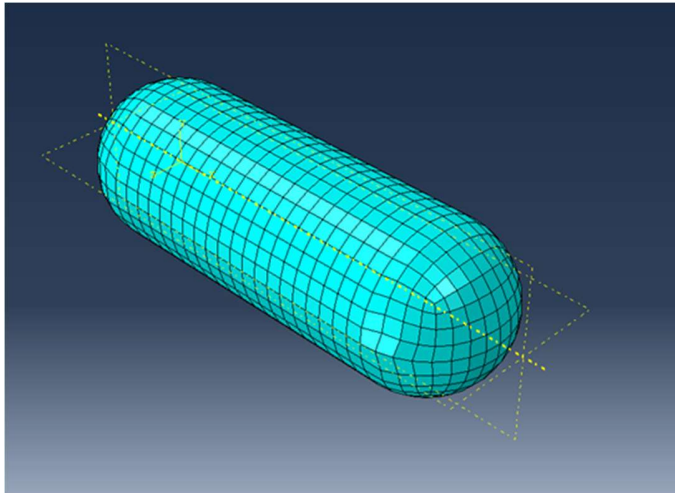
A stress analysis was completed with an internal pressure of 1.21 MPa and no boundary conditions:



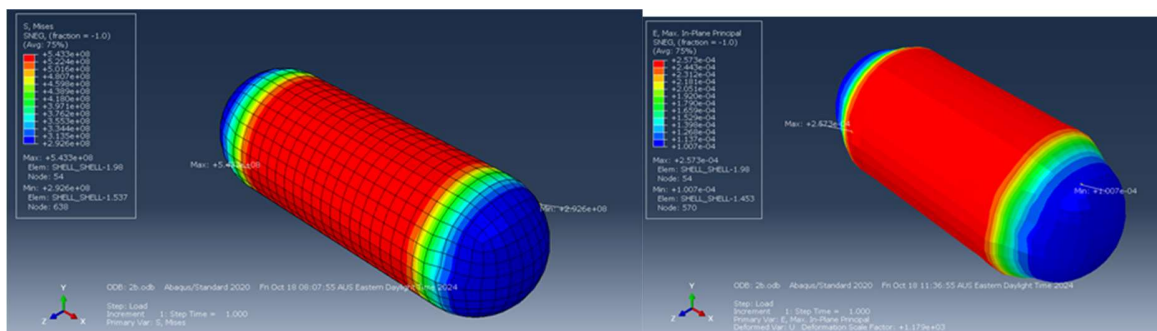
The maximum stress and strain both occur at the same location at the interface between the vessel shell and head with a stress of 57.9 MPa and strain of 2.72×10^{-4} .

b. Analysis using Shell Elements

For the shell element, the part was meshed using quad-dominated elements to balance accuracy of the simulation without causing warping.



A stress analysis was completed with an internal pressure of 1.21 MPa and no boundary conditions:

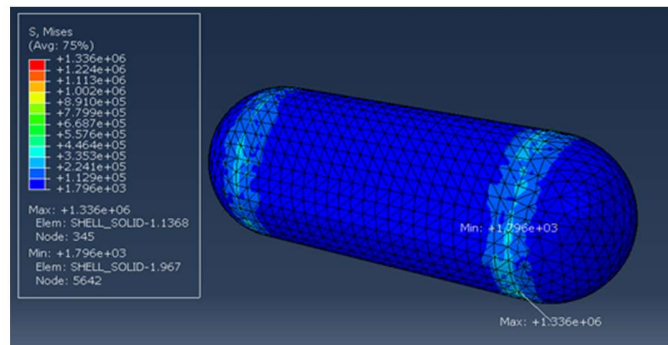
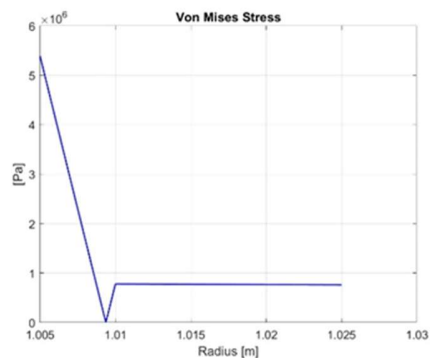


The maximum stress was also located at the interface of the shell and head but with a magnitude of which is within 10% of the tetrahedral mesh. This is to be expected as the thickness-to-radius ratio is less than 1/10.

c. Elasto-Plastic Material Analysis

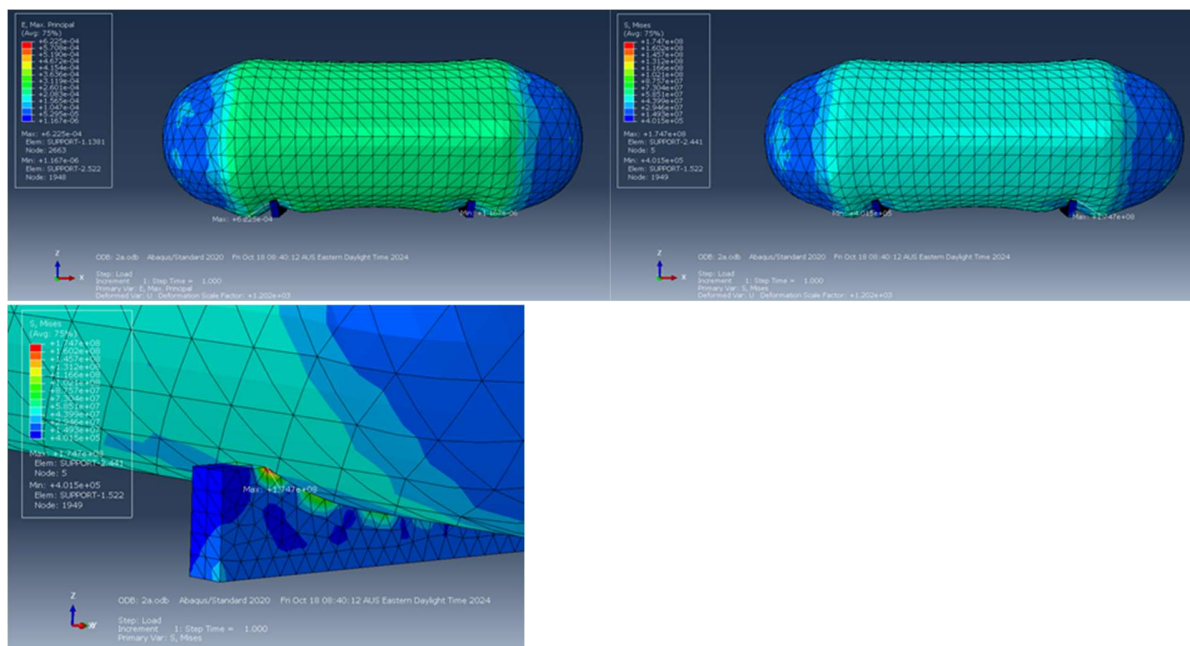
The pressure inside the vessel was set as 12.084 MPa for 1 second, before equalizing with the outer pressure. The tetrahedral mesh was used as this model was a solid, which is able to simulate differences in stress across the thickness of the vessel, unlike the shell part.

The highest residual von-mises stress is 1.34 MPa located at the interface of the head and the shell. According to the previously calculated data the highest von-mises stress is 5.3 MPa. This suggests the assumption that the highest stress occurring at the shell was incorrect, or the mesh was not accurate enough to properly model the elasto-plastic material. It is suggested to repeat calculation with more mesh elements. Also, the Tresca Criteria is considered a conservative criterion which may suggest why the stress is far lower in the numerical solution



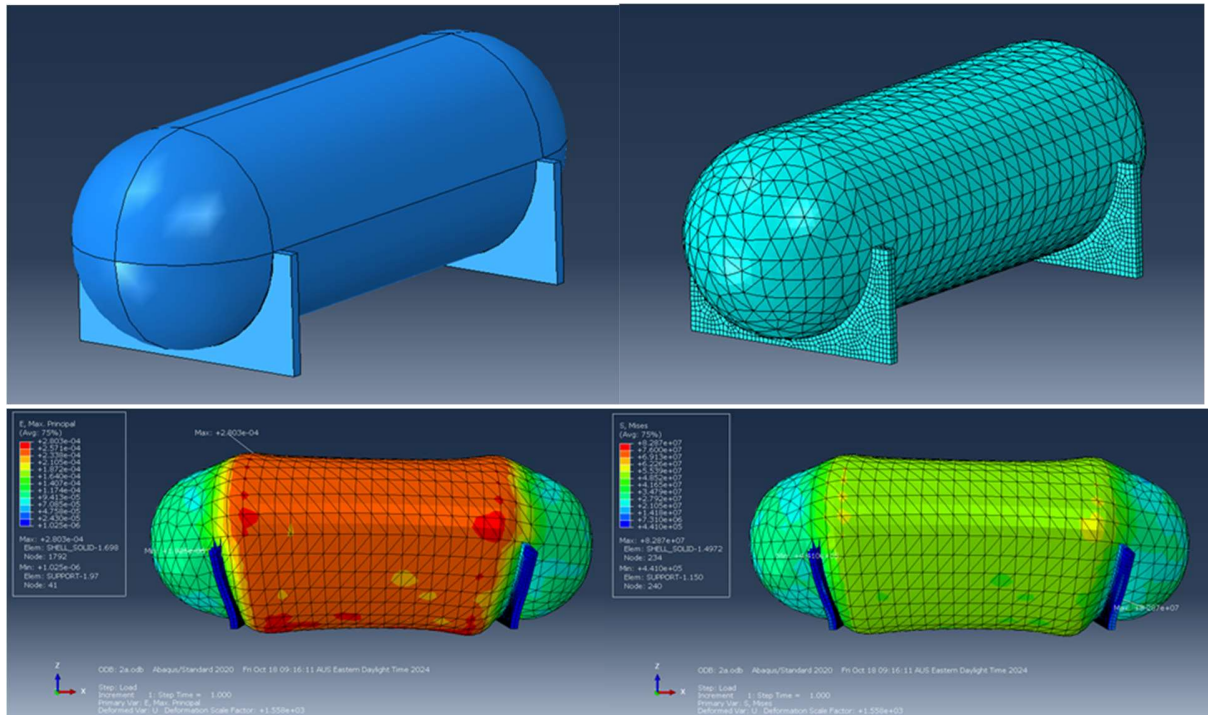
d. Inclusion of Gravitational Effects

To analyse the effect of the supports, gravity was added to the simulation with a 9.8 m/s^2 acceleration in the negative z direction. An encastre boundary condition was added to the base of the supports (assuming perfect join with ground) and a tie contact was added on the interface of the shell and the supports (assuming no slip between parts). The pressure inside the vessel was returned to 1.21 MPa



Analysing the results, a large concentration of pressure was discovered at the corner of the support with a magnitude of 174 MPa which is greater than the yield stress. This suggests the supports are not well designed to distribute the load of the vessel.

e. Evaluation of Support Effects and Design



To redesign the supports it was decided to alter the shape of the supports by extending them to the halfway point of the cylinder. This way, the load would be distributed over the support rather than concentrated at the corner. Also, in order to reduce the maximum stress in the vessel, the supports were moved outwards to the interface of the head and shell, to act as a brace. It was found through several iterations that this location minimized the von Mises Stresses and Strain of the vessel.